



FFI Norwegian Defence
Research Establishment



SAGE: Ergodic Control for Autonomous and Adaptive Inspection of Subsea Infrastructure

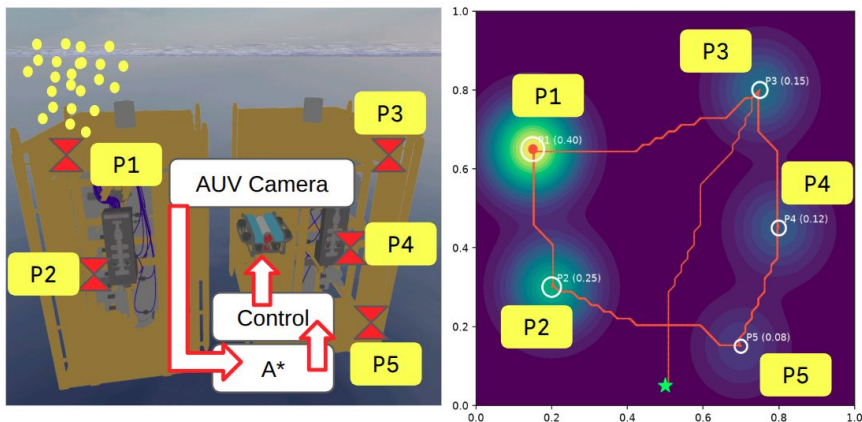
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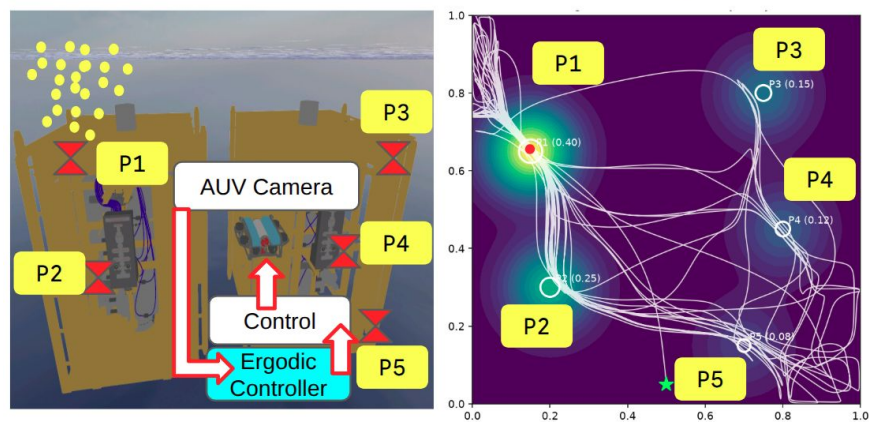
Motivation

Baseline: Planned-Route Loop



Subsea infrastructure inspection often follows **fixed routes** that ignore changing risk conditions.

SAGE (Ours): Autonomous Coverage Loop



SAGE (Semantic and Adaptive Generative Ergodicity) replaces static inspections with adaptive coverage control, dynamically prioritizing inspection time and revisits based on live leak detection.

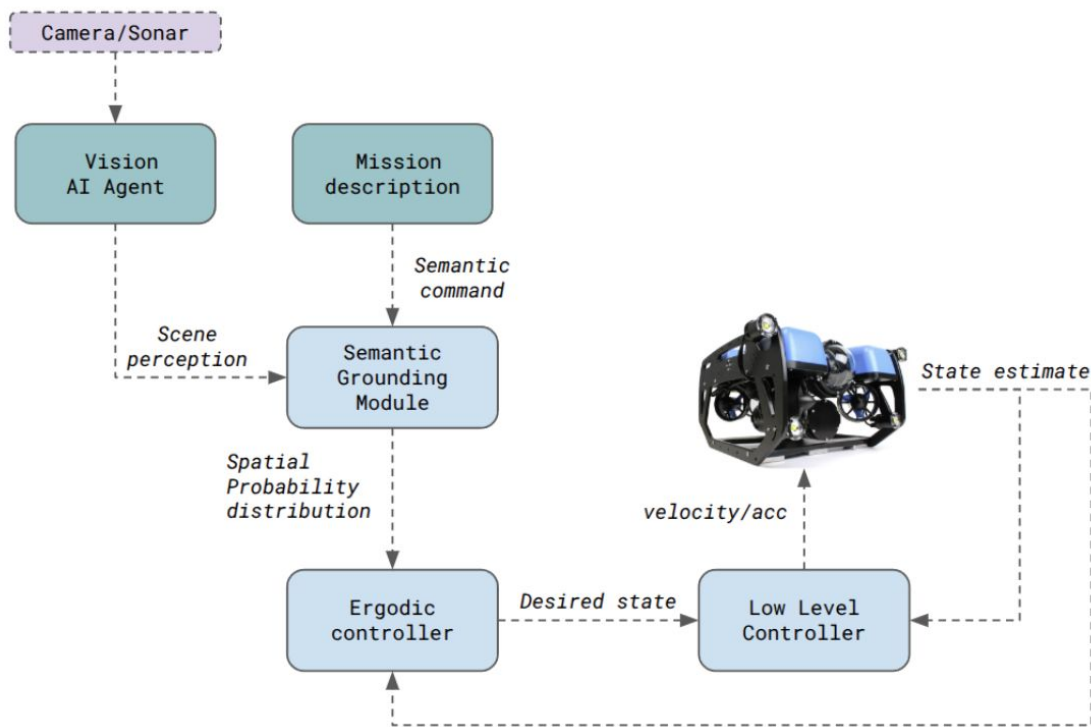
The Architecture & Method

Vision AI Agent: Detects defects and leaks from camera and sonar data.

Semantic Grounding: Builds a live spatial probability map from mission goals and sensor data.

Ergodic Controller: Adapts coverage in real time without route re-planning.

Low-Level Controller: Executes commands using continuous state feedback.



Key Results

Equal Weights (Scenario 1): Both methods cover all 5 valves evenly.

Graded Weights (Scenario 2): Inspection automatically focuses more on higher-risk valves (+0.88 correlation).

Leaking Valve (Scenario 3):

- 1.4× Faster Revisit: 5.8s vs. 8.1s with A*.
- More Coverage: 50 visits vs. 37 with A*.
- Instant Adaptation: Re-routes automatically in real time.

